Eliseo Bao

Ph.D. Student in Computer Science

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scholar.google.com/citations?user=evKkNoMAAAAJ

Research Interests

- **NLP for Healthcare** — detecting and explaining depression-related symptoms in social media text against clinical frameworks such as the DSM-5 and the BDI-II.
- **Explainable AI (XAI)** — interpretable, clinician-facing reasoning rather than opaque classification.
- **Conversational AI** — LLM-based agents and the simulation and evaluation of persona-grounded conversation.
- **Multilingual NLP** — adapting large language models to underrepresented languages such as Galician.

Education

University of A Coruña, Ph.D. in Computer Science – Galicia, Spain 2024 – present

University of A Coruña, M.Sc. in Computer Science – Galicia, Spain 2024
Extraordinary Master's Award, 2023/2024 cohort.

University of A Coruña, B.Sc. in Computer Science – Galicia, Spain 2022

IES San Clemente, Tech. in Application Development – Galicia, Spain 2018

Experience

Predoctoral Fellow, Universidade da Coruña 2024 – present

- Xunta de Galicia Predoctoral Grant ED481A-2024-079.

Visiting Researcher, Natural Language Processing Group, University of Sheffield – Sheffield, UK Oct 2025 – Dec 2025

- Hosted by Dr. Xi Wang; work on LLM persona simulation led to the Eval4Sim paper (CIKM 2026).

Research Assistant, CITIC Centro de Investigacion TIC - Universidade da Coruña 2023 – 2024

- Working at the Information Retrieval Lab (IRLab) within the project PLEC2021-007662 Big-eRisk Early Prediction of Personal Risks on Massive Data funded by MCIN/AEI/10.13039/501100011033 and by the European Union NextGenerationEU/PRTR.

Internship, ITG Instituto Tecnológico de Galicia 2022 – 2023

- Contributed to the CEL.IA group, developing NLP solutions, with a focus on the ASR research line.

Apprenticeship, NATICA, SISTEMAS DE INFORMACION SL 2018 – 2018

Publications

Improving clinical reliability of LLM reasoning for depression assessment via structured generation and GRPO July 2026

Shows GRPO-trained LLMs can explain depression-screening decisions through DSM-5-aligned reasoning without sacrificing classification accuracy.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1177/00368504261467483 (Science Progress, 109(3))

Eval4Sim: An Evaluation Framework for Persona Simulation Nov 2026

Introduces a three-dimensional framework (adherence, consistency, naturalness) for evaluating how well LLM personas simulate human conversation.
Eliseo Bao, Anxo Pérez, Javier Parapar, Xi Wang
10.1145/3799682.3840172 (Proc. of the 35th ACM Int'l Conference on Information and Knowledge Management (CIKM 2026), Rome, Italy. Forthcoming)

Learning Evidence of Depression Symptoms via Prompt Induction July 2026

Proposes Symptom Induction, compressing labeled examples into interpretable guidelines that improve LLM classification of depressive symptoms.
Eliseo Bao, Anxo Pérez, David Otero, Javier Parapar
10.1145/3805712.3809873 (Proc. of the 49th International ACM SIGIR Conference (SIGIR 2026), Melbourne, Australia)

ReDSM5: A Reddit Dataset for DSM-5 Depression Detection Nov 2025

A 1,484-post Reddit dataset with sentence-level, clinician-annotated DSM-5 depressive-symptom labels and rationales.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1145/3746252.3761610 (Proc. of the 34th ACM Int'l Conference on Information and Knowledge Management (CIKM 2025), Seoul, Republic of Korea)

Towards Explainable and Safe Systems for Health Data July 2025

Doctoral consortium paper outlining a research plan for transparent, explainable, and safe health information systems.
Eliseo Bao
10.1145/3726302.3730121 (SIGIR 2025 Doctoral Consortium, Padua, Italy)

How does depression talk on social media? Modeling depression language with relevance-based statistical language models Dec 2025

Builds relevance-based statistical language models of depression-related language to rank users by risk across Reddit and Twitter.
Eliseo Bao, Anxo Pérez, David Otero, Javier Parapar
10.1016/j.osnem.2025.100339 (Online Social Networks and Media, 50, 100339)

MindWell: A Conversational Agent for Professional Depression Screening on Social Media Apr 2025

Demonstrates an open-source RAG chatbot that helps clinicians review social media history for BDI-II depressive symptoms.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1007/978-3-031-88720-8_9 (Advances in Information Retrieval — ECIR 2025, LNCS, Part V, pp. 47-52, Lucca, Italy)

Explainable depression symptom detection in social media Sept 2024

Transformer-based models that both detect and explain depressive symptom markers in social media text.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1007/s13755-024-00303-9 (Health Information Science and Systems, 12(1), 47)

Adapting Large Language Models for Underrepresented Languages 2024

Introduces a Galician-adapted Llama 3.1 model and instruction dataset, outperforming the base model and another Galician model.
Eliseo Bao, Anxo Pérez, Javier Parapar
(VII Congreso XoveTIC, Universidade da Coruña — CICAS-Sngular Award)

MindWell: an open-source chatbot to assist in the detection and monitoring of depressive disorders 2024

Designed and evaluated the MindWell chatbot with a professional psychologist.
Eliseo Bao
(M.Sc. Thesis, Universidade da Coruña)

Ranking of Reddit users using Relevance Models for depressive disorders 2022

Built depression-risk vocabularies and ranking methods from Reddit using relevance-based language models.
Eliseo Bao
(B.Sc. Thesis, Universidade da Coruña)

Awards

CICAS-Sngular Award 2024

Awarded for the best work in line with best practices for Open Science, Open Data, Open Access and Open Source. Certification · Publication

Extraordinary Master's Award 2025

M.Sc. in Computer Engineering, 2023/2024 cohort.

B.Sc. thesis category award 2023

Certification · YouTube video

Finalist, Best Open Source University Project Award 2022

Publication

Talks

- **2026** — "Explainable depression symptom detection in social media," 2nd Symposium on Mental Health and Artificial Intelligence, Université de Caen Normandie, France.

Program Committees & Reviewing

- **2027** — Program Committee, ECIR 2027 (Resource Papers track).
- **2027** — Reviewer, WWW 2027 (full papers).
- **2026** — Program Committee, CIKM 2026 (Resources track; Short Papers track).
- **2026** — Program Committee, CLEF 2026 (eRisk Lab).
- **2026** — Reviewer, SIGIR 2026 (full papers).
- **2026** — Reviewer, PLOS Digital Health.
- **2025** — Program Committee, CIKM 2025 (Resource Papers track).
- **2025** — Program Committee, CLEF 2025 (eRisk: Early Risk Prediction on the Internet).
- **2025** — Reviewer, SIGIR 2025 (subreviewer / external reviewer).
- **2025** — Reviewer, Health Information Science and Systems (Springer), two reviews.
- **2024** — Reviewer, Health Information Science and Systems (Springer).

Languages

Galician
Native

Spanish
Native

English
Professional working proficiency